WATER RESOURCES, WELLS & PUMPS · NOTE WL01

Aquifers and Types of Wells — Dug, Driven, Tube, Bore and Cavity Wells, Filter Points

Course topics: IDE6.2 L5, IDE6.2 L7

Wells and aquifers

- Groundwater is used by building wells and tanks and installing lifts and pumps; tanks can also catch runoff from small areas. Irrigation wells differ from domestic wells because they must deliver large volumes.
- A **well** is a hole, usually vertical, reaching a water-bearing formation; the water drawn from it is its **yield**. Using groundwater efficiently depends on designing the well to suit the formation.
- An **aquifer** is a porous soil or geological formation that yields significant quantities of water — think of it as an underground storage reservoir over a large area.

Aquifer	Also called	Features
Unconfined	Free, phreatic, non-artesian	Lies above an impervious stratum; water not under pressure; the water table undulates with recharge, discharge, pumping and permeability. In a well, static water level = water table
Confined	Artesian, pressure	Lies below an impervious layer; water under pressure, so it rises in a well above the bottom of the confining bed. Recharged where the confining bed rises to the surface or ends underground

Piezometric surface: the imaginary surface of hydrostatic pressure level in a confined aquifer, shown by the water level in a well. If it lies above the ground the well **flows** (artesian); if below, the well is **sub-artesian**.

Classification of wells

Basis	Types
Aquifer tapped	**Gravity (percolation) well** — from an unconfined aquifer; water level = water table. **Artesian well** (called a "cavitation well" in the notes) — from a confined aquifer; water rises to the piezometric level; artesian if above ground, sub-artesian if below
Depth	**Shallow (surface) wells** — pumping water level ≤ 6 m, the practical suction limit of pumps; easily contaminated by percolating water. **Deep wells** — pumping level > 6 m; built on an impervious **mota** layer (clay, cemented sand, kankar or other hard subsoil) and drawing from pervious strata below it through a bore
Construction	Dug, driven and drilled wells

Dug (open) wells

- Dug down to the water-bearing strata, circular or rectangular; circular wells are 1.5 to 6 m in diameter. They usually tap unconfined aquifers, and their size lets them **store** a large volume.
- Dig them by hand with pick-axe and shovel, lifting spoil by bucket, rope and pulley — take care against cave-ins in unstable ground. Large wells may use clam-shell or orange-peel buckets.

- Take irrigation wells 4 to 10 m below the dry-weather water table.
- **Lining (steining):** stone, brick or concrete; make the upper 3 m watertight with mortar, cement plaster or concrete; perforate or leave open the part in the aquifer. Seat the curb firmly at the bottom and raise its top at least 30 cm above ground.
- **Backfill** between casing and excavation with sand and gravel up to the water-bearing stratum, coarser at the bottom; a puddled-clay or concrete fill stops surface water creeping down the casing.
- **Sanitary protection** for pumped wells: a concrete platform with manhole, pump base and pump hole; a tight iron or concrete manhole cover, or a close wire mesh to keep out birds, animals and pollutants.
- **Quicksand:** remove it by bucket or pump while keeping the pit partly full of water; drive temporary or permanent curbing 0.3 to 1.2 m below the excavation bottom; finish with a thin layer of coarse sand and successive gravel layers forming a graded filter.

Dug-cum-bore wells

- Where a confined aquifer lies below a dug well, bore through the impervious bottom to raise the yield — 20–30 m deep in hard rock. In alluvium, line the bore with pipe projecting 30–60 cm above the well bottom to avoid silting. The yield is much higher than an ordinary dug well.
- Geological Survey of India studies: lateral tunnels along rock fissures also raise yield. Bore them with indigenous tools — twisted or pointed bits on 2 cm square rods 1–1.5 m long — keeping the water level below the bore; air-compressor horizontal boring is also used. Deepening a well to reach deeper water-bearing formations increases yield too.

Driven wells

- Drive or jet a pipe with a perforated **well point** (acting as a strainer) into the aquifer; diameter 3 to 10 cm; a standard-weight water pipe with threaded couplings forms the casing.
- Usually shallow, where the water table is shallow and the soil is not hard enough to damage the point; fitted with reciprocating suction pumps. Yields are low — mainly for domestic supply.

Drilled wells (tube wells)

- Made with drilling machines — tube wells are always drilled. Usually deep, often through more than one aquifer: drill past shallow aquifers to tap deeper ones with more water and pressure. A true artesian effect raises the water above the aquifer, reducing drawdown and lift while the pressure lasts.
- Diameter usually 15 to 45 cm; deep tube wells reach 100 m or more; yield in alluvium 30 to 60 L/s.
- Gravel-packed and lined with metal casing: plain pipe through non-water-bearing layers (and usually through the first water-bearing layer), perforated or strainer pipe in the aquifer zone.

Well	Key features
Borewell	Tube well in hard rock; uncased except in the weathered top; about 15 cm diameter, normally 60 m or more deep; yield 3 to 10 L/s with a submersible pump; drinking water and small-scale irrigation
Cavity tube well	No screen — solid pipe only; water enters through a cavity formed at the pipe bottom. Needs a thick clay layer (at least about 3 m) over a sandy aquifer. Cheaper than screened wells, but fails without the clay roof or where the aquifer holds much fine sand, silt or clay
Filter point	Shallow tube well in alluvium, 10 to 15 m deep and 10 to 15 cm in diameter, usually sunk by hand; strainer at the bottom connected straight to the pump suction (no separate suction pipe). Only where the pumping or spring water level is within 7 m of the ground or pump; ideal in sandy river basins with a high water table — Kaveri delta (Tamil Nadu), Krishna and Godavari deltas (Andhra Pradesh)

Developing a cavity well

1. Bore through the whole clay layer into the sand.
2. Remove sand with a sand bailer to start a cavity.
3. Pump with a high-capacity pump, increasing the drawdown gradually until a stable cavity forms ("development of the cavity").
4. Operate the well at a drawdown lower than used in development, to get sand-free water.

ICAR eCourse (Groundwater, Wells and Pumps): filter points are shallow (< 15 m) and small (< 7.5 cm diameter) in deltaic coarse sand and gravel, often worked by hand; cavity wells are also called non-penetrating wells, and they fail mainly when the clay roof collapses.

Types of filter point

Type	Construction
Metal	Pipe 10 to 15 cm; holes 8 to 15 mm at 2.5 to 4 cm spacing with about 20 cm blank at both ends; conical driver point at the bottom, threaded top; holes covered with copper or brass mesh; lasts 5 to 7 years; relatively costly
Coir	Cheap and popular in coastal South India: steel-rod cage with flat steel bands riveted or welded, conical wooden bottom plug, top coupling, wound with 3 to 5 mm coir rope; large filter area; unaffected by alkaline or acidic water
Slotted pipe	Round metal pipe with slots 2.5 to 6.5 mm wide and 5 to 7.5 cm long, spaced 6 to 20 mm slot to slot and 25 to 40 mm row to row; enough open area to avoid high entrance velocity; gravel pack of suitable size and thickness

Sinking method	How
Direct driving	Drive the point by repeated blows, screwing on pipes as it goes, usually to 10–12 m. The aquifer cannot be checked properly and the filter is often damaged, so the method is being discontinued
Casing-pipe method	Lower casing with sand line and bailers to the aquifer, place the filter, withdraw the casing, pack the annulus with pea gravel and connect the pump suction straight to the filter point

Multiple-well systems

- Wells close together have overlapping cones of depression, which increases the drawdown in each — place each well outside the others' area of influence.

- Where the water table is shallow, in deltas or in thin aquifers, a battery of wells can be cheaper than one deep well — in isolated groups or continuous patterns, their suction lines joined to a common header feeding one pump.
- In waterlogged areas such systems help control the water table.

Sources: ICAR eCourse 'Groundwater, Wells and Pumps' — Dr M. K. Jha (IIT Kharagpur), reviewed by Dr P. B. S. Sarma (IARI New Delhi). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. An aquifer lying below an impervious layer, with water under pressure, is

(a) unconfined (b) confined
(c) perched (d) phreatic

Q2. A shallow well is one whose pumping water level is within

(a) 2 m (b) 6 m
(c) 15 m (d) 30 m

Q3. Tube wells in hard-rock areas, uncased except at the top, are called

(a) cavity wells (b) borewells
(c) filter points (d) driven wells

Q4. A cavity well needs a clay layer at least about

(a) 0.3 m thick (b) 3 m thick
(c) 30 m thick (d) 100 m thick

Q5. If the piezometric surface lies above the ground, the well is

(a) sub-artesian (b) flowing artesian
(c) gravity (d) dry

Q6. Coir filter points are popular in

(a) Punjab (b) coastal South India
(c) western Rajasthan (d) Himalayan valleys

Answers

Q1 — (b) confined
Q2 — (b) 6 m — *The practical suction limit of pumps.*
Q3 — (b) borewells
Q4 — (b) 3 m thick
Q5 — (b) flowing artesian
Q6 — (b) coastal South India